DATA ANALYST PORTFOLIO CASE STUDY | SYNTHETIC DATASET

Are Invoice Collections Deteriorating?

Northstar Workflow | MySQL invoice and payment analysis | January 2024 – December 2025

TOTAL INVOICES

214,599

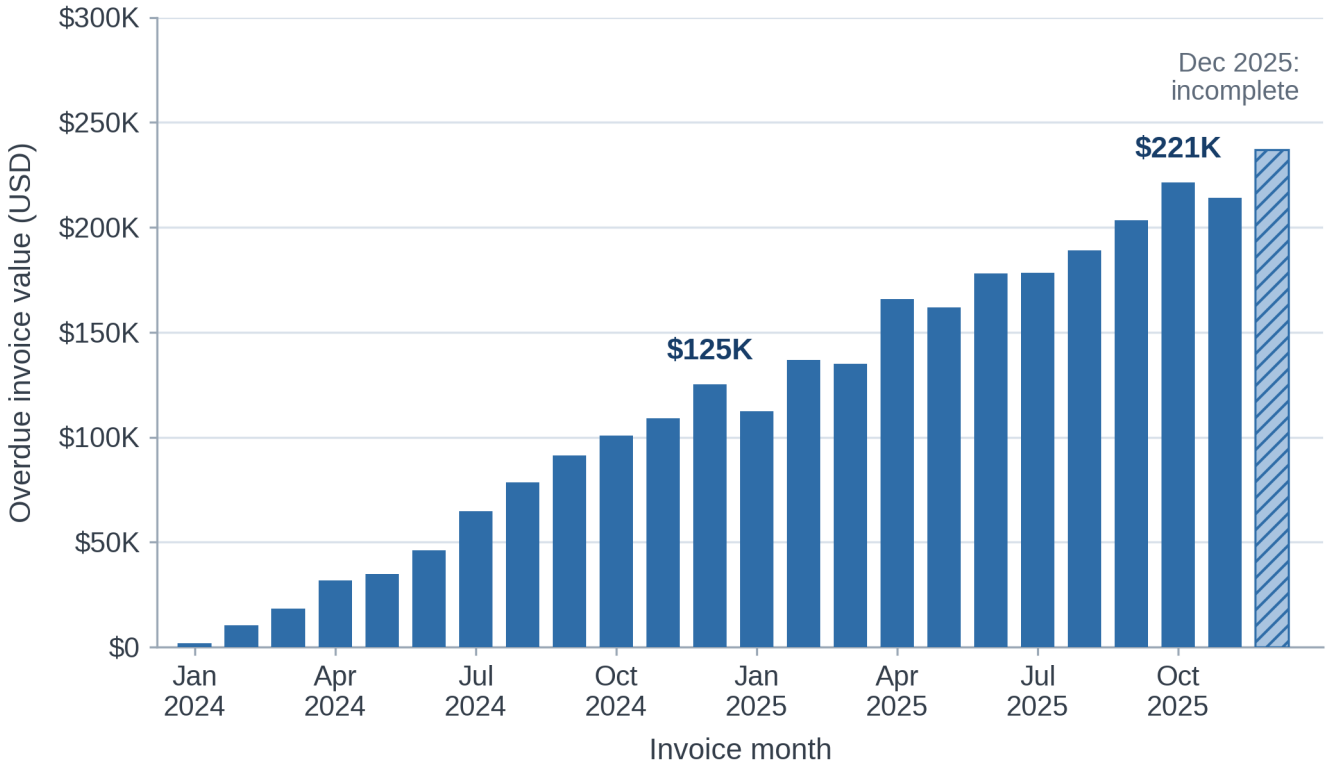
OVERDUE INVOICES

13,043 6.1% of all invoices

OVERDUE INVOICE VALUE

~$2.85M $2,846,303.00 exact

Monthly overdue invoice value



Invoice-month cohorts. December 2025 (hatched) is incomplete: invoice-date coverage ends on 5 December 2025, so it is not comparable with full months.

Key finding

Overdue value rose; the overdue share did not.

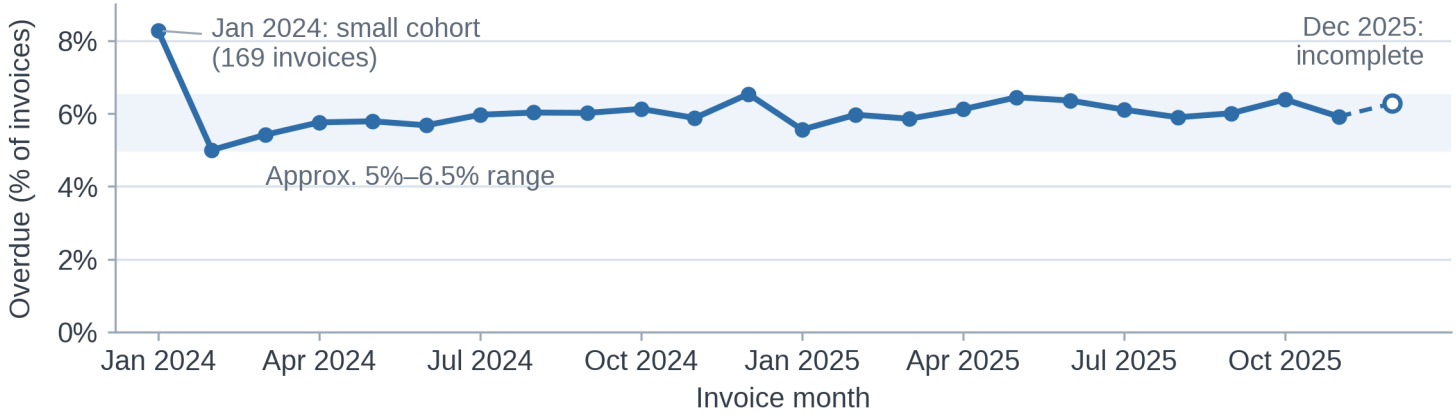
Overdue value grew with invoice volume, passing $200,000 in several late-2025 months. But most monthly cohorts stayed near 5%–6.5% overdue, and recorded payment turnaround held near 8.5 days.

The evidence **does not establish clear deterioration in collection performance**. Rising overdue value still matters: it is a larger outstanding exposure.

Business question: are invoice collections deteriorating, and what does the evidence say about overdue invoice exposure?

What Does the Evidence Tell Us?

Monthly overdue invoices as a percentage of all invoices, plus recorded payment turnaround



AVG. RECORDED PAYMENT TURNAROUND

~8.5 days

Covers only the 201,555 invoices with recorded payments; it says nothing about unpaid invoices.

Note: percentages are calculated by invoice month using the currently recorded invoice status. They are not historical month-end overdue rates.

Key findings

- **Overdue exposure grew:** overdue value increased as invoice volumes grew.
- **Overdue proportion broadly stable:** generally around 5%–6.5% across monthly invoice cohorts.
- **Recorded turnaround stable:** about 8.5 days among invoices with recorded payments.

Recommendations

1. **Prioritise high-value overdue balances.** Review the largest invoices marked overdue, verify balances and payment status, and follow up confirmed unpaid amounts.
2. **Monitor overdue value alongside the overdue rate** each month, to separate growth in exposure from changes in the share of invoices marked overdue.
3. **Improve invoice tracking.** Capture invoice due dates and keep month-end records of balances and statuses, to support overdue-ageing analysis and more reliable monitoring.

Limitations

- Synthetic dataset, not real client or employer data.
- No invoice due dates, so time overdue cannot be measured.
- No month-end snapshots (statuses and balances saved at each month-end), so results are by invoice-month cohort, not month-end balances.
- 60 invoice amounts are missing (2 overdue); overdue value may be slightly understated.
- December 2025 is incomplete (invoice dates end 5 Dec 2025).
- The 8.5-day turnaround excludes unpaid invoices and does not describe them.

Conclusion: Overdue value increased alongside invoice volume, but the overdue proportion remained broadly stable. The evidence does not establish clear deterioration in collection performance; continued monitoring is warranted.